ADITYA UNIVERSITY

Database management system

Cornerstone Project

University Academic Management System

Presented by

E.SIVAMANI (26B21AI294)
G.MANIKANTA KARTHEEK (26B21AI313)
S.MANIDEEP REDDY (26B21AIB74)
N.SREENATH REDDY (26B21AI795)

GUIDED BY :

Mrs G.L.NarasambaVanguri, M.tech, Ph.D

Department of

Artificial Intelligence and Machine Learning

ADITYA UNIVERSITY

Contents

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Entity
6. Attributes

ADITYA UNIVERSITY

ABSTRACT

The **University Academic Management System** is a database project developed using **SQL and MySQL Workbench** to manage university academic information in an organized manner. The system stores important data related to **students, departments, faculty, courses, enrollment, examinations, results, attendance, classrooms, and timetables**.

The main purpose of this project is to make academic data easier to store, update, retrieve, and manage. The system uses **tables, primary keys, foreign keys, and relationships** to maintain accurate and organized data.

This project provides a simple and efficient way to manage university academic records using a relational database.

Introduction

➢**SQL (Structured Query Language)** is a standard language used to create, store, retrieve, update, and manage data in relational databases. SQL is used to perform operations such as creating tables, inserting data, updating records, deleting records, and retrieving information.

In our project, **MySQL Workbench** is used to design and manage the database for the **University Academic Management System**.

The system is designed to store and manage academic information such as **Students, Departments, Faculty, Courses, Enrollment, Examinations, Results, Attendance, Classrooms, and Timetables**.

➢The project uses tables and relationships to organize the academic data. **Primary Keys and Foreign Keys** are used to identify records and establish relationships between tables.

The main purpose of our project is to provide an organized database for managing university academic information and to make storing and retrieving data easier and more efficient.

ADITYA UNIVERSITY

Problem Statement

Universities manage a large amount of academic information such as student details, departments, faculty, courses, enrollment, examinations, results, attendance, classrooms, and timetables. Managing this information manually or using separate records can make it difficult to store, update, and retrieve data accurately.

There may be problems such as **duplicate records, data errors, difficulty in finding information, and lack of proper relationships between academic records**.

Therefore, our project **University Academic Management System** aims to provide a structured database using **SQL and MySQL Workbench** to store and manage university academic information efficiently and make data retrieval easier.

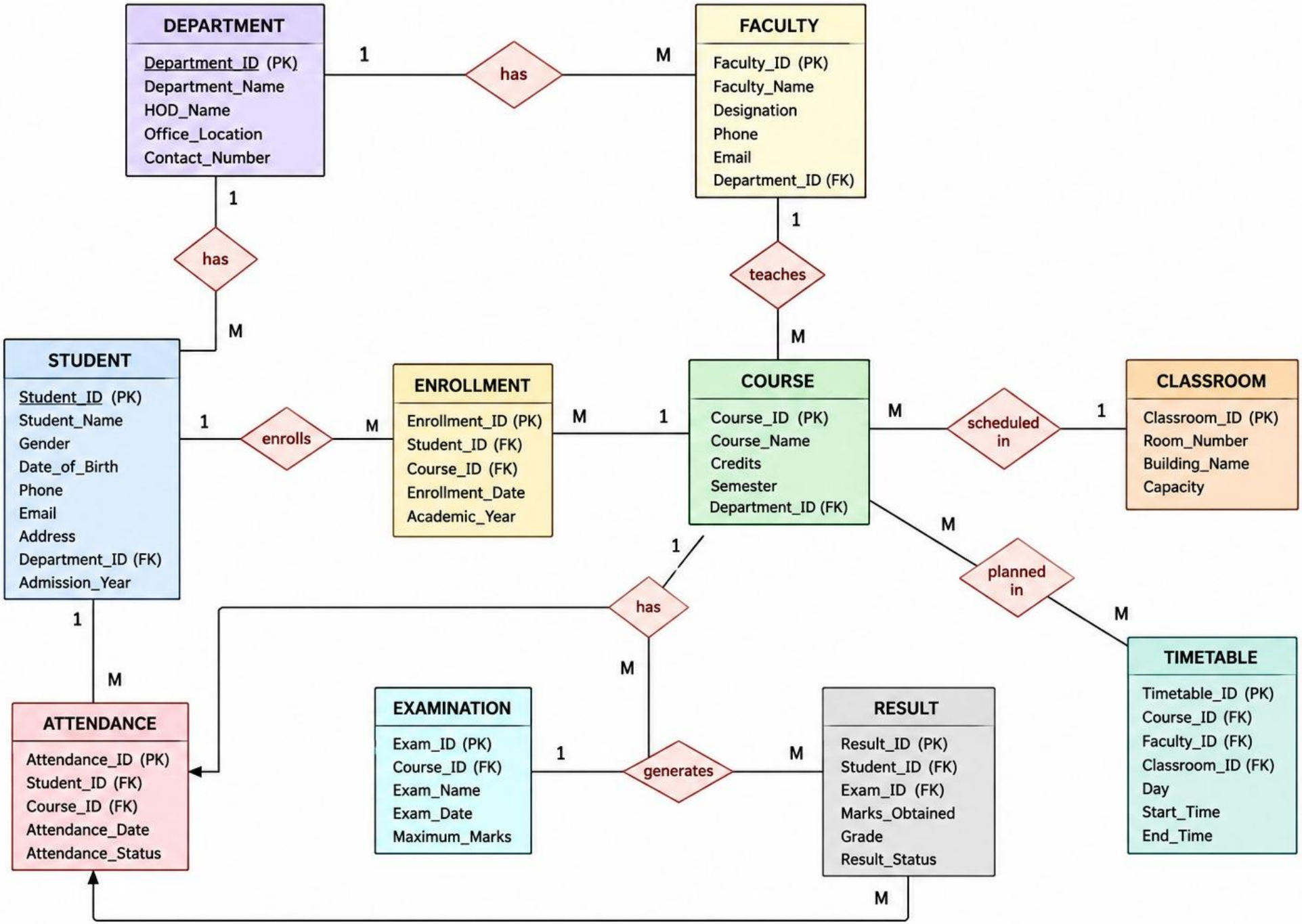
ADITYA UNIVERSITY

Objectives

1. To develop a structured database for storing and managing university academic information.
2. To maintain complete academic records of Students, Departments, Faculty, Courses, Enrollment, Examinations, Results, Attendance, Classrooms, and Timetables.
3. To establish relationships between different academic entities using appropriate database relationships.
4. To use Primary Keys and Foreign Keys to uniquely identify records and maintain connections between related tables.
5. To use SQL and MySQL Workbench for efficient storage, organization, and management of academic data.
6. To enable easy data operations such as inserting, updating, deleting, and retrieving academic records.
7. To maintain data accuracy, consistency, and integrity throughout the database.
8. To improve the efficiency and accessibility of academic information for authorized users.
9. To improve data security and consistency by using a centralized database.

ADITYA UNIVERSITY

Entity



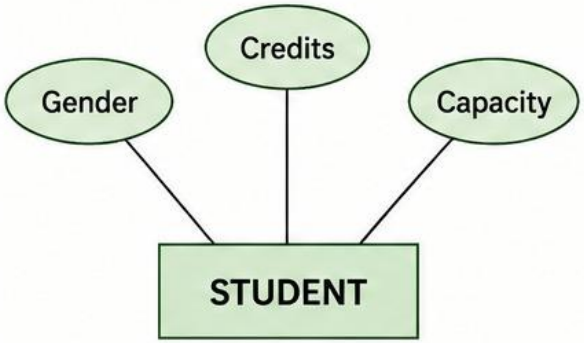
ADITYA UNIVERSITY

Attributes

1. SIMPLE ATTRIBUTE

Cannot be divided into smaller meaningful parts.

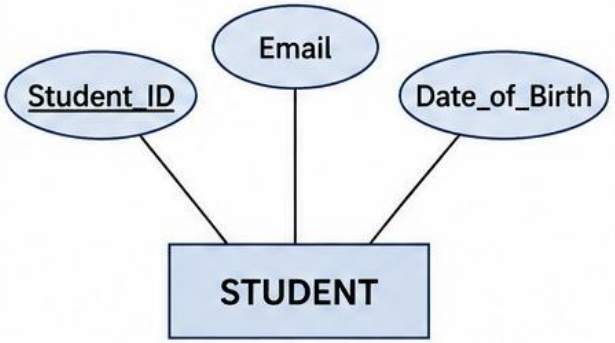
Example: Gender, Credits, Capacity



2. SINGLE-VALUED ATTRIBUTE

Has only one value for each entity.

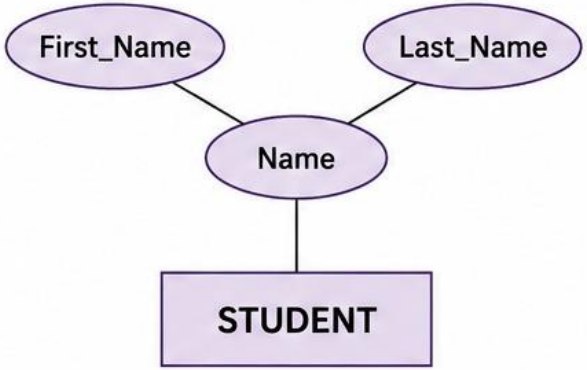
Example: Student_ID, Email, Date_of_Birth



3. COMPOSITE ATTRIBUTE

Can be divided into smaller attributes.

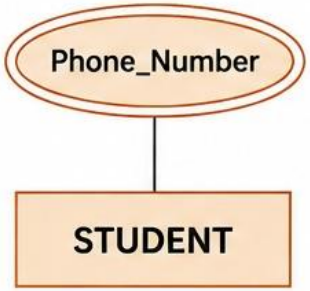
Example: Name → First_Name + Last_Name



4. MULTIVALUED ATTRIBUTE

Can have multiple values for one entity.

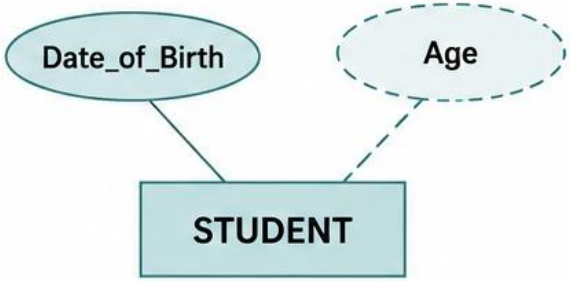
Example: Phone_Number if a student can have multiple phone numbers.

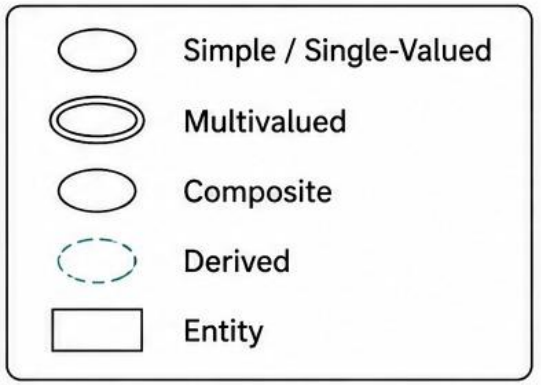


5. DERIVED ATTRIBUTE

Can be calculated from another attribute.

Example: Age can be derived from Date_of_Birth.





ADITYA UNIVERSITY

THANK YOU!